EXPERIMENT 10

AIM: **Student Management System Using ReactJS with Routing**

SOLUTION:

React Router is used to implement navigation in React applications without reloading the page.

Procedure

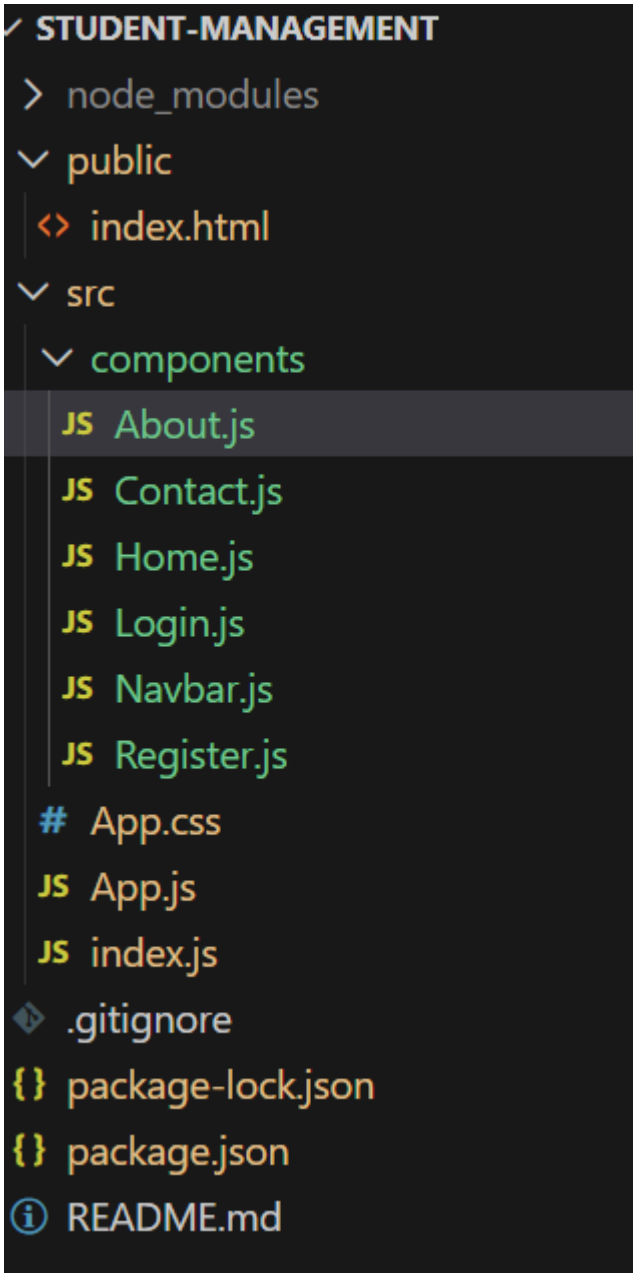
Step 1: Create React Application

npx create-react-app student-management

Step 2: Navigate to Project Folder

cd student-management

Student-management Folder Structure:



Program:

public/index.html

```
<!DOCTYPE html>
```

```html
<html lang="en">
  <head>
    <title>Student Management System</title>
  </head>
  <body>
    <noscript>You need to enable JavaScript to run this app.</noscript>
    <div id="root"></div>
  </body>
</html>
```

src/components/About.js

```
import React from "react";
function About() {
  return (
    <div style={{ textAlign: "center" }}>
      <h2>About</h2>
      <p>Student Management System built using React.</p>
    </div>
  );
}
export default About;
```

src/components/Contact.js

```
import React from "react";
function Contact() {
  return (
    <div style={{ textAlign: "center" }}>
      <h2>Contact Us</h2>
      <p>Email: support@student.com</p>
      <p>Phone: 9876543210</p>
    </div>
  );
}
export default Contact;
```

src/components/Home.js

```
import React from "react";
```

```
function Home() {
  return <h2 style={{textAlign:"center"}}>Welcome to Student Management System</h2>;
}
export default Home;
```

src/components/Login.js

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```
import React from "react";
function Login() {
  return (
    <div style={{ textAlign: "center" }}>
      <h2>Student Login</h2>
      <input type="email" placeholder="Email" /><br /><br />
      <input type="password" placeholder="Password" /><br /><br />
      <button>Login</button>
    </div>
  );
}
export default Login;
```

src/components/Navbar.js

```
import React from "react";
import { Link } from "react-router-dom";
import "../App.css";   // make sure CSS is imported
function Navbar() {
  return (
    <nav className="navStyle">
      <div className="menuStyle">
        <Link to="/" className="linkStyle">Home</Link>
        <Link to="/register" className="linkStyle">Register</Link>
        <Link to="/login" className="linkStyle">Login</Link>
        <Link to="/contact" className="linkStyle">Contact</Link>
        <Link to="/about" className="linkStyle">About</Link>
      </div>
    </nav>
  );
}
```

```
export default Navbar;
```

src/components/Register.js

```
import React from "react";
function Register() {
  return (
    <div style={{ textAlign: "center" }}>
      <h2>Student Registration</h2>
      <input type="text" placeholder="Name" /><br /><br />
      <input type="email" placeholder="Email" /><br /><br />
      <input type="password" placeholder="Password" /><br /><br />
      <button>Register</button>
    </div>
  );
}
export default Register;
```

src/index.js

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

src/App.js

```
import React from "react";
import { BrowserRouter as Router, Routes, Route } from "react-router-dom";
import Navbar from "./components/Navbar";
import Home from "./components/Home";
import Register from "./components/Register";
import Login from "./components/Login";
import Contact from "./components/Contact";
import About from "./components/About";
function App() {
  return (
```

```
    <Router>
      <Navbar />
      <div style={{ padding: "20px" }}>
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/register" element={<Register />} />
          <Route path="/login" element={<Login />} />
          <Route path="/contact" element={<Contact />} />
          <Route path="/about" element={<About />} />
        </Routes>
      </div>
    </Router>
  );
}
export default App;
```

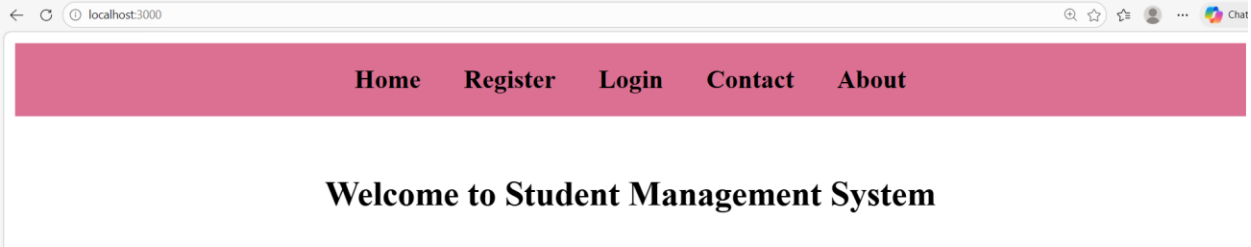
src/App.css

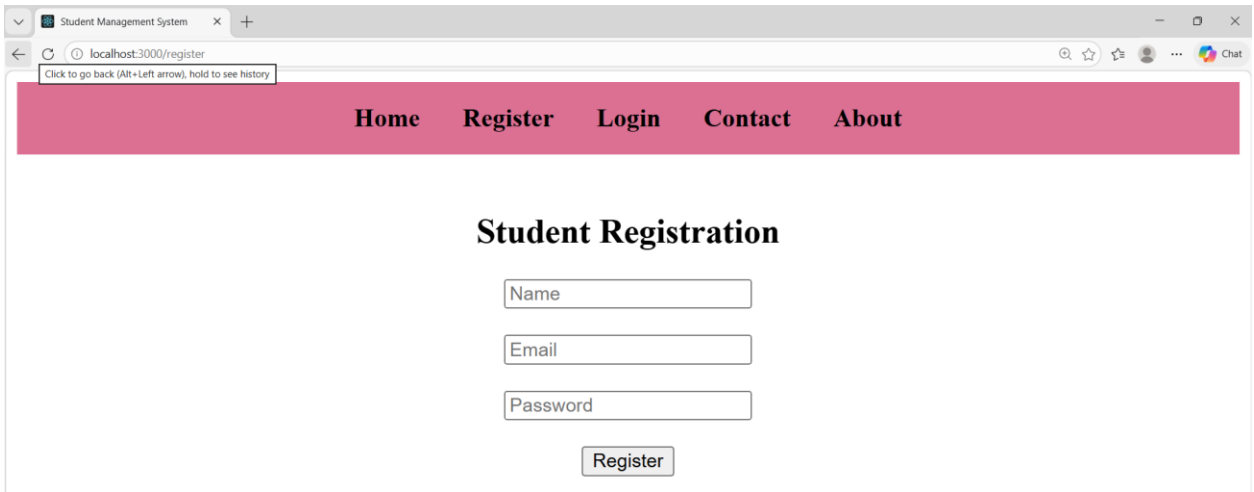
```
.navStyle {
  background-color:palevioletred;
  padding: 15px;
}
.menuStyle {
  display: flex;
  justify-content: center;
  gap: 30px;
}
.linkStyle {
  text-decoration: none;
  color: black;
  font-weight: bold;
  font-size: 18px;
}
```

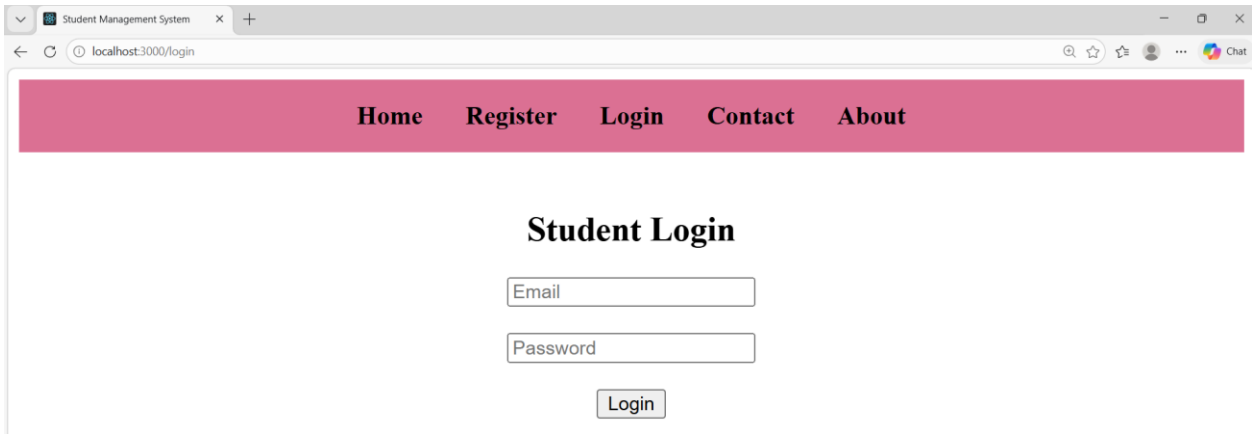
Step 3: Start Application

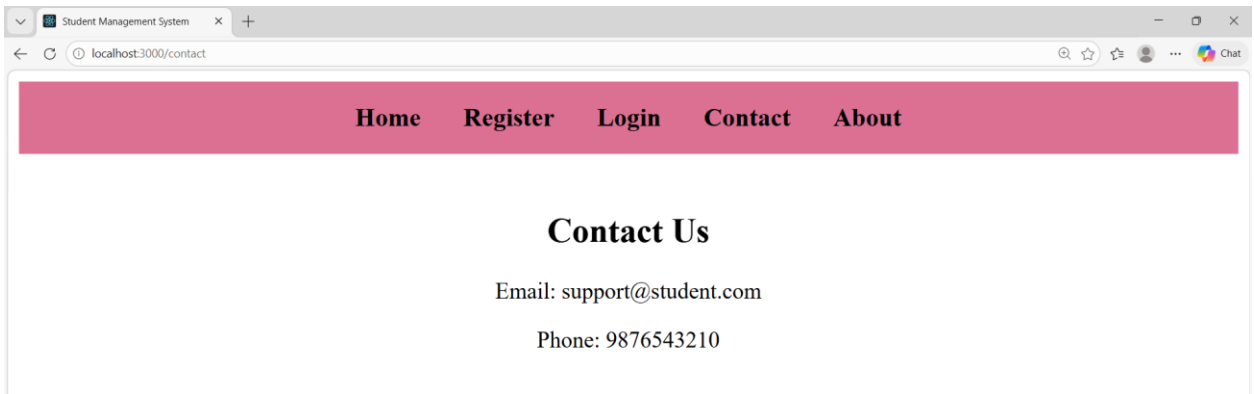
```
>npm start
```

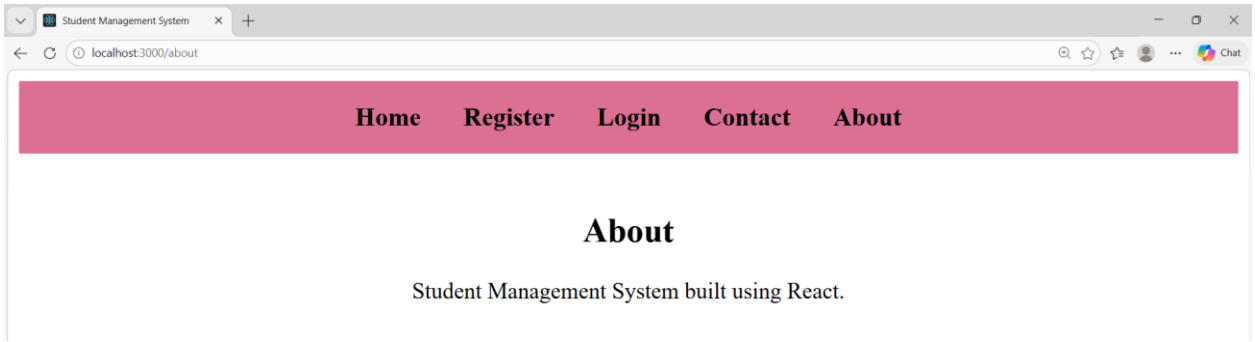
Output:


localhost:3000
Home Register Login Contact About
Welcome to Student Management System


Student Management System
localhost:3000/register
Click to go back (Alt+Left arrow), hold to see history
Home Register Login Contact About
Student Registration
Name
Email
Password
Register


Student Management System
localhost:3000/login
Home Register Login Contact About
Student Login
Email
Password
Login


Student Management System
localhost:3000/contact
Home Register Login Contact About
Contact Us
Email: support@student.com
Phone: 9876543210


Student Management System
localhost:3000/about
Chat
Home
Register
Login
Contact
About
About
Student Management System built using React.